THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of a triangle is 54 cm^2 and its base is 12 cm. Its height (cm) is: 1

(a) 10 **(b)** 8 **(c)** 11 **(d)** 9

2. The simplest form of $\frac{18}{42}$ is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{4}{7}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{3}{7}$

3. A rectangle ABCD has AB = 14 cm. M is the midpoint of AB. AM (cm) is: 1

(a) 21 **(b)** 6 **(c)** 8 **(d)** 7

4. A wheel has 8 equal spokes. The smallest angle of rotation that fits the wheel back on itself is (degrees): 1

(a) 75 **(b)** 35 **(c)** 45 **(d)** 55

5. The number of integers between -6 and 5 (not counting them) that are greater than 0 is: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

6. A rectangle 12 cm by 9 cm is cut along a diagonal into two triangles. The perimeter of the rectangle is (cm): 1

(a) 30 **(b)** 51 **(c)** 42 **(d)** 54

7. $\frac{3}{8}$ of 56 m is (in m): 1

(a) 23 **(b)** 22 **(c)** 21 **(d)** 20

8. Two circles have radii 9 cm and 4 cm and their centres are 13 cm apart. The circles: 1

(a) touch externally **(b)** do not touch (separate)

(c) one lies inside the other **(d)** touch internally

9. A regular polygon has each angle of rotation $18°$. The number of lines of symmetry is: 1

(a) 20 **(b)** 21 **(c)** 22 **(d)** 19

10. 22 + (-30) + 5 = 1

(a) -1 **(b)** 7 **(c)** -5 **(d)** -3

11. How many square metres are there in a rectangle of 14 m by 11 m? 1

(a) 155 **(b)** 154 **(c)** 168 **(d)** 153

12. $4\frac{3}{4} - 1\frac{5}{6}$ = 1

(a) 5 **(b)** $3\frac{3}{4}$ **(c)** $4\frac{7}{12}$ **(d)** $2\frac{11}{12}$

13. In rectangle ABCD, AB = 10 cm and BC = 7 cm. AC is a diagonal. The perimeter of rectangle (cm) is: 1

(a) 34 **(b)** 41 **(c)** 44 **(d)** 24

14. The total angle through which a regular octagon turns when rotated 5 times through its smallest angle (degrees): 1

(a) 255 **(b)** 215 **(c)** 235 **(d)** 225

15. -8 - 5 - (-3) = 1

(a) -8 **(b)** -12 **(c)** 0 **(d)** -10

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A square has side 10 cm. Its side is increased by 2 cm. Find the increase in area. 2

17. Arrange $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{8}$ in descending order. 2

18. Find the area of a rectangle constructed with sides 10 cm and 7 cm and the area of a square of side 8 cm. Which is larger? 2

19. Draw (describe) the reflection of the triangle with vertices at x = 1, 3, 2 (heights 0, 0, 3) in a vertical mirror line at x = 0. Give the new x-coordinates. 2

20. Write all integers between -5 and 2 in ascending order. How many are there? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the value of $6 - (-4) + (-9) - 5$. 3

22. A rectangular lawn 40 m by 25 m has a path 3 m wide inside it all around. Find the area of the path. 3

23. A farmer ploughed $\frac{2}{5}$ of a field on Monday and $\frac{1}{4}$ of it on Tuesday. What part of the field remains to be ploughed? 3

24. A square of side 8 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

OR

A regular polygon has 6 lines of symmetry. Find its angle of rotation, the angle after 4 turns and the number of turns to make a full rotation. 3

25. A train station has platforms numbered with 0 at ground level. A lift moves 6 floors below ground, then 2 floors up, then 7 floors below again. Find its final floor and the total number of floors moved. 3

OR

Two rectangles have the same perimeter 48 cm: one is 18 cm by 6 cm and the other is 10 cm by 14 cm. Which has the larger area and by how much? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Given figure types: a square (lines 4) and a regular octagon (lines 8). Which has the larger angle of rotation? Find both angles, their difference and say which can fit into the other's symmetry (explain: 4 divides 8 or not). 5

OR

The temperatures (°C) at 5 places are -8, 3, -2, 11, -5. Arrange them from the coldest to the hottest, find the difference between the hottest and the coldest places and the sum of all temperatures. 5

27. A rectangular hall 12 m by 9 m is to be covered with square tiles of side 40 cm costing ₹25 each. Find the number of tiles and the total cost. 5

OR

Ravi had ₹240. He spent $\frac{1}{6}$ of it on books and $\frac{2}{5}$ of the rest on a bag. How much does he have left? 5

END OF QUESTION PAPER **Total: 50 Marks**

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